

STORMSHIELD

PROFESSIONAL SERVICES

NABACKUPTOJSON USER GUIDE

v2.2.0.0

Historique de révision

Version	Date	Description
2.2.0.0	13/11/2026	Document Initialization

Table des matières

WELCOME 4
- Products concerned 4
- Recommendations on the operating environment 4
- Conventions used in this document 4

INSTALLATION 5
- Installing the .NET 10 Desktop Runtime 5
- Installing the application 5
- Using the portable version 6
- Settings location 7
- Uninstalling the application 7
- Error messages 7

GETTING STARTED 8
- Understanding the graphical interface 8
- Home module 8
- Keyboard shortcuts 9
- Error messages 10

CONVERSION 11
- Source backup 11
- Converting a backup 12
- Exporting the JSON file 12
- Structure of the JSON file 13
- Error messages 13

CONTENT EXPLORATION 14
- Archive content 14
- Detail of the selected file 15
- Error messages 16

COMPARISON 17
- Comparing two backups 17
- Viewing the differences 18
- Exporting and reopening a diff file 19
- Error messages 20

Stormshield copyright 2026

SETTINGS 21
Decryption tool 21
Appearance and language 22
Supported formats 22
About 22
Error messages 22
COMMAND LINE 23
Converting a backup 23
Comparing two backups 23
Options 23
Error messages 24

WELCOME

Welcome to the naBackupToJson user guide.

This guide details the features of the application's modules, and provides the information you need to convert and compare configuration backups of Stormshield Network Security (SNS) firewalls.

naBackupToJson allows you to:

- **Convert** a configuration backup (`.na` file) into a single, structured JSON file,
- **Explore** the content of a backup before exporting it,
- **Compare** two backups to identify additions, removals and changes,
- **Automate** these operations from the command line.

Products concerned

Configuration backups (`.na`) of Stormshield Network Security firewalls, decoded by the `decbackup.exe` tool.

Workstations: Windows 10 and Windows 11, 64-bit.

Recommendations on the operating environment

A configuration backup contains the whole security policy of a firewall: network objects, filtering and NAT rules, VPN configuration, accounts and certificates. It must be handled as confidential data.

- naBackupToJson processes backups only on the workstation. No data is sent over the network.
- Temporary files created during decoding are deleted at the end of each operation, whether it succeeds or not.
- Backup passwords are never saved nor written to the log.
- The JSON and diff files produced contain the configuration in clear text: store them with the same level of protection as the original backups.

> `decbackup.exe` is not included with naBackupToJson. Use your own copy of this tool. For more information, refer to the INSTALLATION section.

Conventions used in this document

- Names of buttons, fields and options of the interface are shown in **bold**.
- Navigation paths are shown as **Settings** > **About**.
- File names, commands and values to type are shown in `monospace font`.
- The screenshots in this document were taken from an anonymized backup.

INSTALLATION

naBackupToJson is delivered in two forms:

- **Installer** (`naBackupToJson-<version>-setup-win-x64.exe`): installs the application on the workstation, with a Start menu shortcut,
- **Portable version** (`naBackupToJson-<version>-portable-win-x64.zip`): runs without installation, for example from a USB drive.

Both forms contain the graphical application `naBackupToJson.exe` and the command-line tool `naBackupToJson-cli.exe` .

Installing the .NET 10 Desktop Runtime

The graphical application `naBackupToJson.exe` runs on Microsoft's **.NET 10 Desktop Runtime (x64)**, installed once on the workstation. The command-line tool `naBackupToJson-cli.exe` does not need it.

To check whether it is present, type `dotnet --list-runtimes` in a terminal: a `Microsoft.WindowsDesktop.App 10.…` line means it is installed.

To install the .NET 10 Desktop Runtime:

1. Open the Download .NET 10.0 page,
2. In the **.NET Desktop Runtime** section, download the **Windows x64** installer,
3. Run it and follow the instructions.

> Installing the .NET Desktop Runtime requires administrator rights. On a managed workstation, ask your IT department.

Installing the application

To install naBackupToJson:

1. Run `naBackupToJson-<version>-setup-win-x64.exe` ,
2. Choose an install for yourself only (default, no administrator rights) or for all users of the workstation,
3. If naBackupToJson is already installed, choose the action to perform:
   - **Update to version …** (or **Reinstall version …** if the same version is installed),
   - **Uninstall**: starts the uninstallation of the installed version,
   - **Cancel**.

4. Optionally select the additional tasks:
   - **Create a desktop shortcut**,
   - **Open .na backups with naBackupToJson**: associates `.na` files with the application.
5. Confirm the installation by clicking **Install**,
6. Copy your `decbackup.exe` into the installation folder, or set its location in the **Settings** module.

If the .NET 10 Desktop Runtime is missing, the last page of the installer says so and offers to open its download page instead of starting the application. Install it, then start naBackupToJson from the Start menu.

Default installation folders:

Installation type	Folder
For yourself only	`%LOCALAPPDATA%\Programs\naBackupToJson`
For all users	`C:\Program Files\naBackupToJson`

> If the installed version is newer than the installer's, only **Uninstall** and **Cancel** are offered, to avoid going back to an older version by mistake.

> In silent mode (`/VERYSILENT`), no question is asked: the installed version is updated.

Using the portable version

To use the portable version:

1. Extract `naBackupToJson-<version>-portable-win-x64.zip` into a writable folder,
2. Copy your `decbackup.exe` into this folder,
3. Run `naBackupToJson.exe` .

In the portable version, the `naBackupToJson.portable` file in the folder makes the application save its settings and recent files in the `settings.json` file of that same folder. No file is created in the user's Windows profile.

> If the folder is not writable, the application works normally and saves its settings in the Windows profile (`%APPDATA%\naBackupToJson`). The mode in use is shown in **Settings** > **About**.

Settings location

Form	Settings file
Installer	`%APPDATA%\naBackupToJson\settings.json`
Portable version	`settings.json`, in the application folder

The settings file contains the location of `decbackup.exe`, the theme, the language and the recent files list. It contains no password. Uninstalling keeps this file.

Uninstalling the application

To uninstall naBackupToJson:

1. Open **Windows Settings** > **Apps** > **Installed apps**,
2. Select **naBackupToJson**, then click **Uninstall**.

You can also run the installer again and choose **Uninstall**.

Error messages

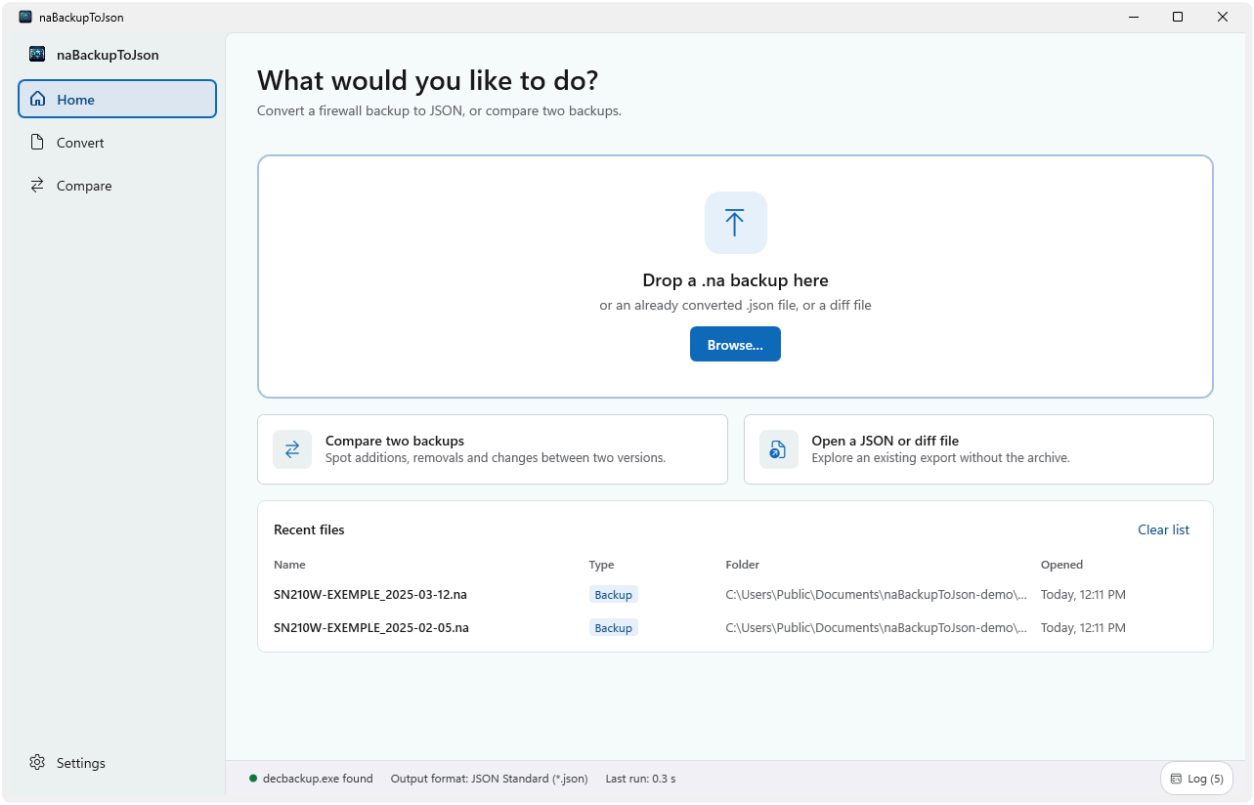
Message or symptom	Solution
At startup, a window says: `You must install or update .NET to run this application.`	The .NET 10 Desktop Runtime is not installed. Click **Yes** to open its download page, or follow Installing the .NET 10 Desktop Runtime.
Status bar: `decbackup.exe not found`	Copy `decbackup.exe` next to `naBackupToJson.exe`, or set its location in **Settings** > **Path to decbackup.exe**.
`A newer version of naBackupToJson (…) is already installed.`	Keep the installed version, or click **Uninstall** and run the installer again.
Copy of `decbackup.exe` denied in `C:\Program Files\naBackupToJson`	This folder requires administrator rights: set the location of `decbackup.exe` in the **Settings** module instead.
The portable version does not show **Portable mode**	The folder is read-only or the `naBackupToJson.portable` file is missing. Extract the zip into a writable folder.
Windows shows "Windows protected your PC"	Click **More info**, then **Run anyway**.

GETTING STARTED

Understanding the graphical interface

The window is made up of 4 areas:

1. The left menu giving access to the modules: **Home**, **Convert**, **Compare** and **Settings**,
2. The active window of the selected module,
3. The status bar, at the bottom of the window, showing the availability of `decbackup.exe`, the output format and the duration of the last operation,
4. The log, shown by clicking **Log** in the status bar, listing the operations performed.



Home module

The Home module is made up of three parts:

- **Drop area**: opens a file by drag and drop or by clicking **Browse...**,
- **Shortcuts**: **Compare two backups** opens the Compare module; **Open a JSON or diff file** opens a file already produced by the application,
- **Recent files**: lists the last ten opened or exported files.

Opening a file

A file dropped anywhere on the window opens in the appropriate module:

File	Module opened
`.na` backup	**Convert**, with the backup selected
JSON file produced by the application	**Convert**, in direct exploration
Diff file	**Compare**, with the differences loaded

Recent files

Column	Description
Name	File name. Click the row to reopen the file.
Type	**Backup**, **JSON** or **Diff**.
Folder	Folder containing the file.
Opened	Date and time of the last opening.

To empty the list, click **Clear list**.

> Passwords are never kept with recent files. To reconvert an encrypted backup, type its password again.

Keyboard shortcuts

Shortcut	Action
`Ctrl+O`	Open a file in the active module
`Ctrl+S`	Export the result of the active module
`Ctrl+F`	Move the cursor to the search field
`Esc`	Clear the search field
`F7` / `Shift+F7`	Show the next / previous difference (Compare module)

Error messages

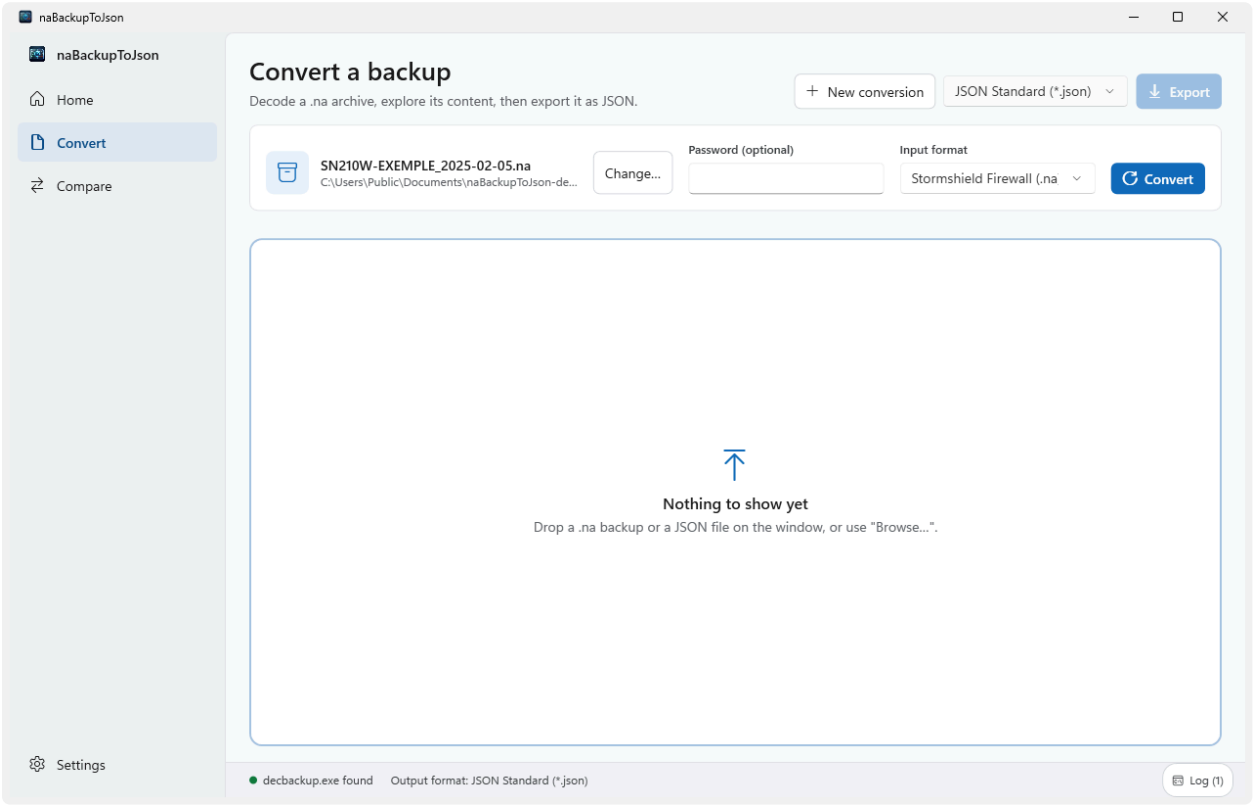
Message	Solution
`decbackup.exe not found` (orange banner on Home)	Click **Configure** and set the location of `decbackup.exe`.
`File not found — "…" cannot be found and was removed from the list.`	The file was moved or deleted: reopen it by clicking **Browse…**.

CONVERSION

This module allows you to decode a configuration backup (`.na`) and export it as a single JSON file.

The Convert module is made up of three parts:

- **Source backup**: selection of the backup, the password and the input format,
- **Archive content**: tree of the backup's files. For more information, refer to the CONTENT EXPLORATION section,
- **Detail**: content of the selected file.



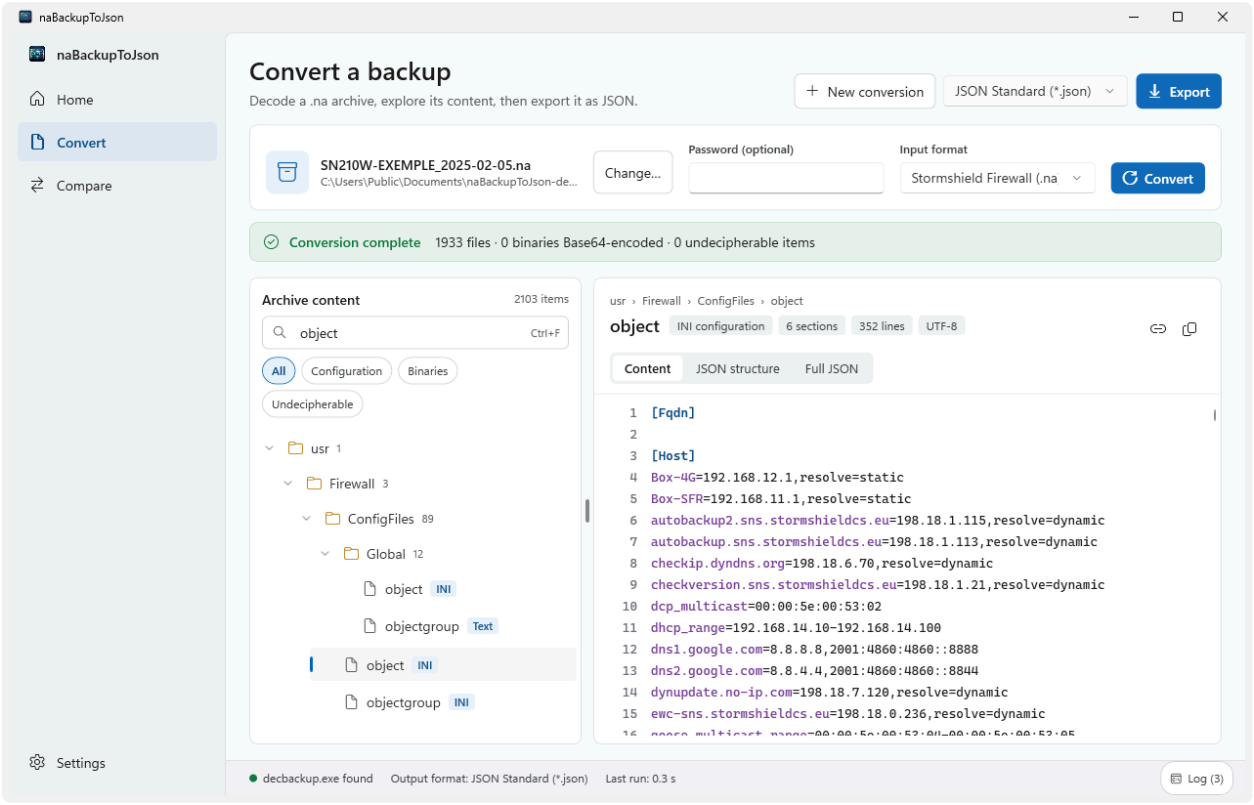
Source backup

Field	Description
Backup name	Name, folder, size and modification date of the selected file. The **Change...** button selects another file.
Password (optional)	Password of the backup, if it was encrypted when created on the firewall.
Input format	Format of the backup. Default value: **Stormshield Firewall (.na)**.

Converting a backup

To convert a backup:

1. Click **Browse…** (or `Ctrl+O`) and select the `.na` file, or drop it on the window,
2. If the backup is encrypted, type its **Password**,
3. Click **Convert**.
   The current step is shown (**Decoding the archive…**, then **Analyzing files…**). The **Cancel** button stops the processing.
4. Read the **Conversion complete** banner, which shows the number of extracted files, of Base64-encoded binary files and of undecipherable items.



Exporting the JSON file

To export the result of the conversion:

1. Check the output format in the list to the left of the **Export** button (default value: **JSON Standard (*.json)**),
2. Click **Export** (or `Ctrl+S`),
3. Choose the location of the file. The suggested name is the backup name, with the `.json` extension.

The **Export complete** banner shows the location of the saved file.

To convert another backup, click **New conversion**.

Structure of the JSON file

Backup item	Representation in the JSON
Backup	Root key named after the `.na` file.
Folder	Key containing its files and folders.
Configuration file (`[Section]` sections)	Object in which each section contains its key/value pairs.
Other text file	List of the file's lines, without comments (`#`) or empty lines.
Binary file	Base64-encoded content.
Unreadable value	`Undecipherable` (`Non déchiffrable` when the application is in French).

Keys are sorted in natural order: `ethernet2` comes before `ethernet10`.

Error messages

Message	Solution
`decbackup.exe not found — Set the location of decbackup.exe in the settings.`	Click **Configure** in the banner and set the location of `decbackup.exe`.
`Decoding failed — Check the password and that the file is a valid backup.`	Type the password again. The detail shown below the message (for example `decrypt and/or verify backup failed (8)`) comes from `decbackup.exe`.
`Invalid archive`	The decoded file is not a readable archive: use a new backup.
`File not accessible`	Close the file in other applications and check your rights on the export folder.
`Conversion cancelled`	Click **Convert** again; no file was produced.

CONTENT EXPLORATION

After a conversion, or when opening a JSON file produced by the application, the Convert module lets you browse the content of the backup before exporting it.

Opening a JSON file requires neither a password nor `decbackup.exe`.

Archive content

The tree shows the folders and files of the backup. The total number of items is shown to the right of the **Archive content** title, and the number of items of each folder is shown to the right of its name.

Each file has a badge indicating its type:

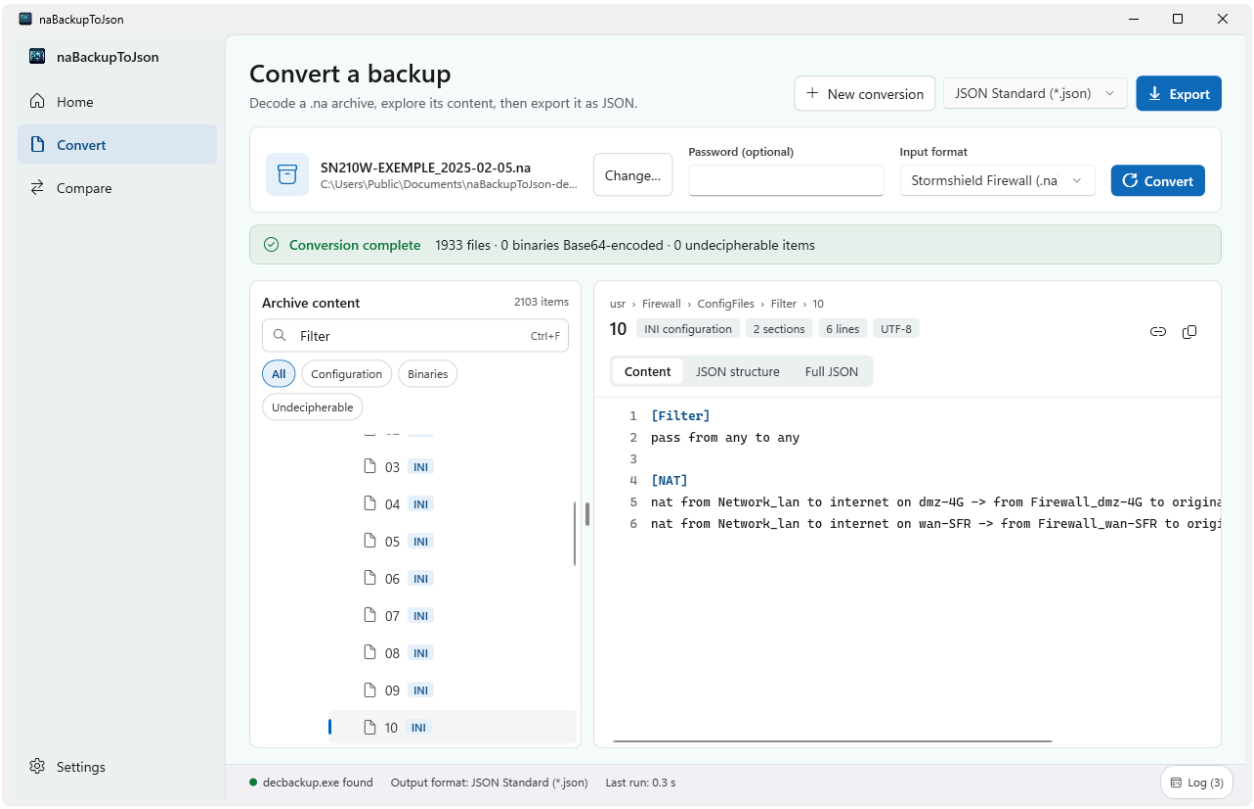
Badge	Description
INI	Configuration file organized in `[Section]` sections.
Text	Text file, represented by the list of its lines.
Base64	Binary file, Base64-encoded.
Undecipherable	File containing at least one unreadable value.

Searching for a file

Type part of the name in the **Search a file…** field (or `Ctrl+F`). Only matching items and their parent folders remain visible. The `Esc` key clears the field.

Filtering by type

The **All**, **Configuration**, **Binaries** and **Undecipherable** filters limit the tree to the chosen file type.



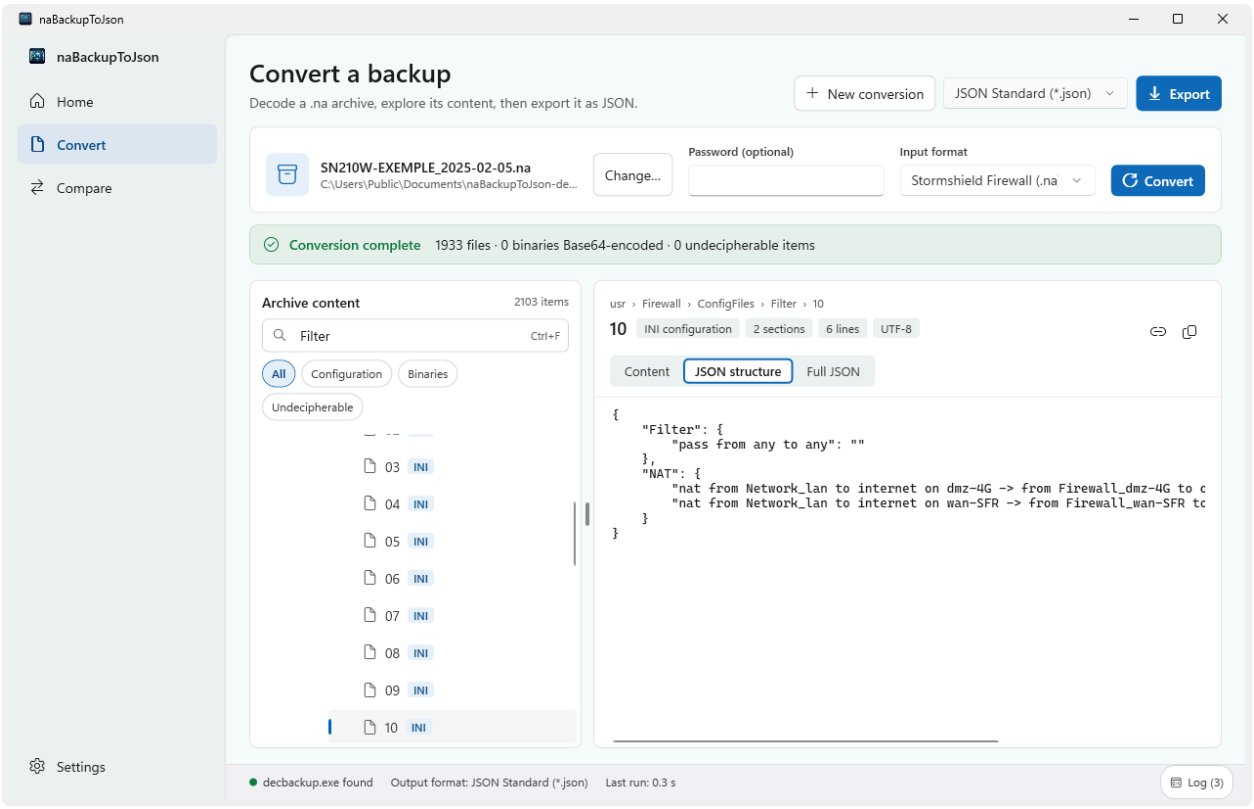
Detail of the selected file

The right part of the module shows:

- the path of the file (for example **usr** > **Firewall** > **ConfigFiles** > **Filter**),
- its name and characteristics: type, number of sections, number of lines, encoding,
- two buttons: **Copy JSON path** (for example `$["backup.na"]["usr"]["Firewall"]`) and **Copy content**.

Three display modes are available:

Mode	Description
Content	Readable view of the file: highlighted sections, colored keys and values, numbered lines.
JSON structure	JSON representation of the selected item.
Full JSON	JSON representation of the whole backup.



Very large contents are truncated on screen with the mention `… [Preview truncated to stay responsive]`. The exported file always contains all the data.

Error messages

Message or symptom	Solution
`No matching item.`	Click **Clear search**, or select the **All** filter.
`Invalid JSON`	The file is not valid JSON: open a file produced by the application.
A binary file shows no content	The content of a binary file is Base64-encoded: select the **JSON structure** mode to show the encoded value.

COMPARISON

This module allows you to compare two backups and identify the items added, removed and modified between them. Each backup can be a `.na` archive or a JSON file produced by the application, in any combination.

The Compare module is made up of three parts:

- **Before** and **After**: selection of the two backups to compare,
- **Differences tree**: items that changed between the two backups,
- **Detail**: removed and added lines for the selected item.

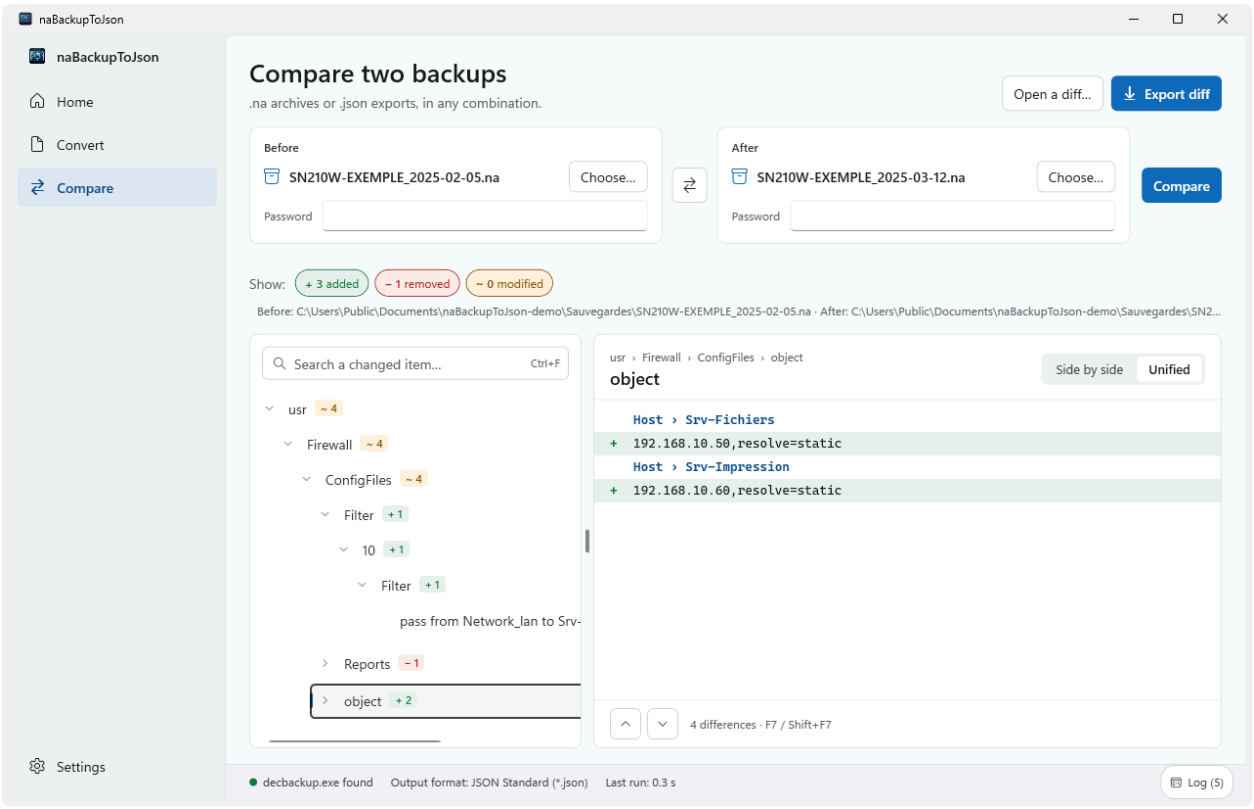
Comparing two backups

To compare two backups:

1. In the **Before** card, click **Choose…** and select the older backup, or drop the file on the card,
2. In the **After** card, select the more recent backup in the same way,
3. If a `.na` backup is encrypted, type its **Password** in the corresponding card,
4. Click **Compare**.

The button between the two cards (**Swap before and after**) exchanges the two backups.

> Two files dropped together on the module fill both cards, the older one in **Before**. A file dropped outside the cards fills the first empty card. A diff file dropped on the module is opened directly.



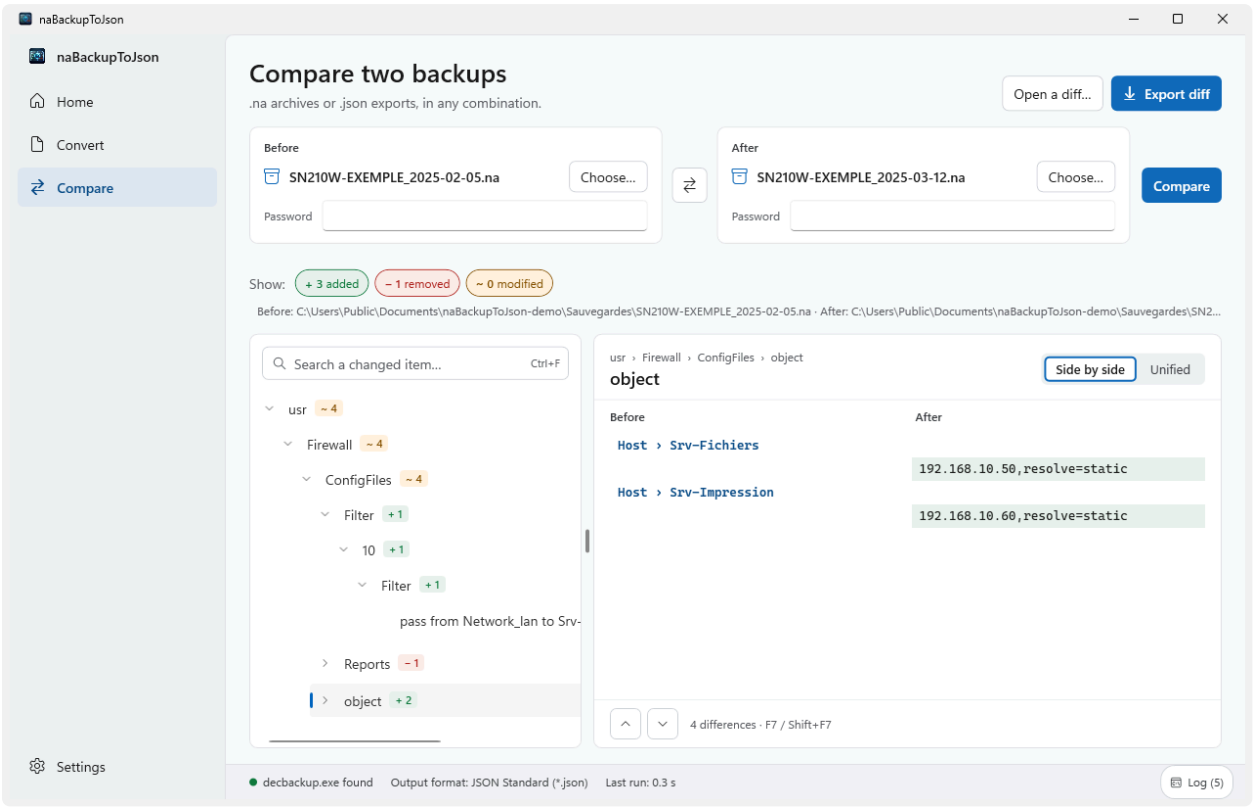
Viewing the differences

The counters below the cards show the number of changes per type. They also act as filters: click a counter to hide or show the corresponding type of change.

Marker	Meaning
`+` (green)	Item added in the **After** backup.
`−` (red)	Item removed from the **After** backup.
`~` (orange)	Item modified. On a folder, indicates that it contains changes.

To view a difference:

1. Select an item in the tree, or type its name in the **Search a changed item…** field,
2. Choose the detail display mode:
   - **Unified**: removed and added lines one below the other,
   - **Side by side**: **Before** version on the left, **After** version on the right.
3. Use the arrows at the bottom of the detail, or the `F7` and `Shift+F7` keys, to show the next or previous difference.



> Large modified content (a binary file, for example) is not copied into the diff file: only the lengths before and after are given.

Exporting and reopening a diff file

To export the result of the comparison:

1. Click **Export diff** (or `Ctrl+S`),
2. Choose the location of the file. The suggested name is `diff_result.json`.

To view a diff file already exported, click **Open a diff…**, or drop the file on the window.

The diff file contains:

Key	Description
`meta`	Compared files, backup names and summary (`added`, `removed`, `modified`).
`diff`	Items that changed. Each item has the `__diff__` key with the value `added`, `removed` or `modified`.

Error messages

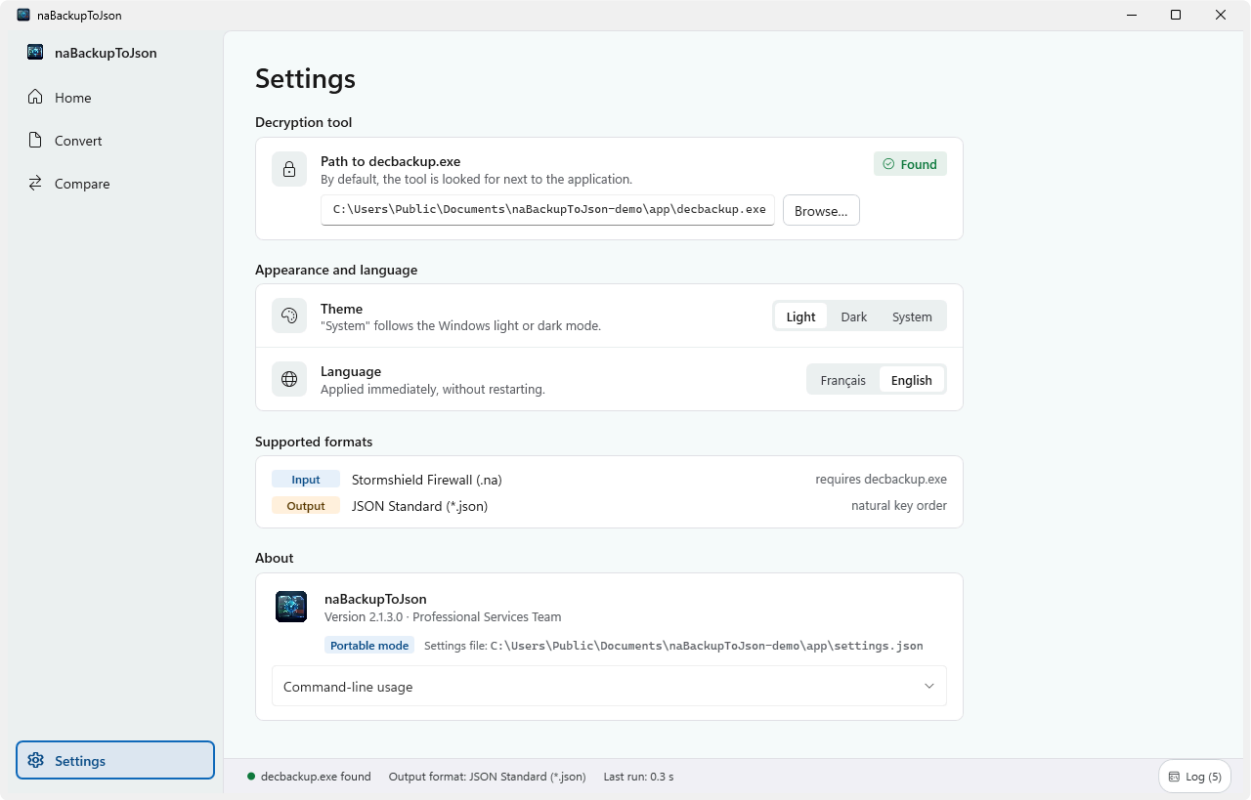
Message	Solution
`Both backups are identical`	No difference was detected: check that you selected the right backups.
`Invalid diff file`	The file does not contain the `meta` and `diff` keys. If it is a backup export, open it from the **Convert** module.
`Decoding failed`	Check the password of the backup concerned (**Before** or **After**).
`decbackup.exe not found`	Click **Configure**, or compare JSON files, which do not require `decbackup.exe` .

SETTINGS

This module allows you to configure the application. Each change is applied and saved immediately, without a confirmation button.

The Settings module is made up of four parts:

- **Decryption tool**: location of `decbackup.exe`,
- **Appearance and language**: theme and language of the interface,
- **Supported formats**: available input and output formats,
- **About**: application version, operating mode and command-line help.



Decryption tool

Field	Description
Path to decbackup.exe	Location of `decbackup.exe`. Type the path or click **Browse...**. Leave the field empty to use the `decbackup.exe` placed next to the application.
Status	**Found**: the tool is available. **Not found**: the path does not match any file.

Appearance and language

Field	Description
Theme	**Light**, **Dark** or **System**. The **System** theme follows the Windows light or dark mode.
Language	**Français** or **English**. The language is applied without restarting.

> The selected language also sets the value written for unreadable items in the next conversions: `Undecipherable` in English, `Non déchiffrable` in French.

Supported formats

Direction	Format	Note
Input	Stormshield Firewall (.na)	Requires `decbackup.exe`.
Output	JSON Standard (*.json)	Keys sorted in natural order.

About

This part shows:

- the application version,
- the **Portable mode** badge, when the application runs as the portable version,
- the location of the **Settings file**,
- the command-line help, by expanding **Command-line usage**. For more information, refer to the COMMAND LINE section.

Error messages

Message or symptom	Solution
Not found badge	The path does not match any file: fix it, or clear the field to use the `decbackup.exe` placed next to the application.
Settings are not kept	Check write access to the folder of the settings file. An unreadable settings file is saved as `settings.json.bak`, then replaced by the default values.

COMMAND LINE

The `naBackupToJson-cli.exe` tool allows you to automate the conversion and comparison of backups from a terminal or a script. It uses the same settings as the graphical application (location of `decbackup.exe`, language).

The exit code is `0` on success and `1` on error.

Converting a backup

To convert a backup:

1. Open a terminal in the application folder,
2. Run the following command:

```
.\naBackupToJson-cli.exe -i C:\Backups\fw.na -o C:\Exports\fw.json -p "password" -l EN
```

Without the `-o` option, the JSON file is saved next to the backup, with the `.json` extension.

Comparing two backups

To compare two backups, run the following command:

```
.\naBackupToJson-cli.exe -db before.na -da after.na -do diff.json -pwb "before pwd" -pwa "after pwd"
```

Without the `-do` option, the `diff_result.json` file is saved in the current folder. The summary of the differences is shown at the end of the processing.

Options

Option	Description
`-i`, `--input`	Backup to convert (`.na`).
`-o`, `--output`	Output file.
`-p`, `--password`	Backup password.
`-d`, `--decbackup`	Location of `decbackup.exe`.
`-ip`, `--input-plugin`	Input format: `Stormshield Firewall (.na)`.

Option	Description
`-op` , `--output-plugin`	Output format: `JSON Standard (*.json)` .
`-l` , `--lang`	Language of messages: `FR` or `EN` .
`-db` , `--diff-before`	"Before" backup (`.json` or `.na`).
`-da` , `--diff-after`	"After" backup (`.json` or `.na`).
`-do` , `--diff-output`	Diff file.
`-pwb` , `--diff-password-before`	Password of the "before" backup.
`-pwa` , `--diff-password-after`	Password of the "after" backup.
`-h` , `--help`	Shows the full help.

To check the exit code in a PowerShell script:

```
if ($LASTEXITCODE -ne 0) { Write-Error "The conversion failed." }
```

⚠ A password passed as an argument may be kept in the terminal history. Clear this history on a shared workstation.

Error messages

Message	Solution
`[-] The path to decbackup.exe is not configured or could not be found. Use the -d option or the application settings.`	Add the `-d C:\path\decbackup.exe` option.
`[-] Error: Execution of decbackup.exe failed. Please check the source file and password if applicable.`	Check the `-p` option (or `-pwb` / `-pwa`). The `Details:` line repeats the `decbackup.exe` message.
`[-] Invalid option or missing value: -l`	Give the option a value. The `-l` option accepts `FR` or `EN` .
`[-] Plugin not found: '…'.`	Use the exact name shown by `--help` or in **Settings** > **Supported formats**.



STORMSHIELD

STORMSHIELD

www.stormshield.com | proserv@stormshield.com

